

ICG ASSIGNMENT 2

Problem

We are building a model which will take satellite image and help us to identify number of buildings and their location in that image. Identification of buildings will be done with the help of yolo. By finding the location of building we can find the shortest path and estimate time to reach our destination.

Data

To train our model we are using **Spacenet_2_Paris_buildings** dataset from kaggle. This dataset contains total 1148 satellite images of Paris buildings along with their coordinates.

Split	Image	Lables
Train	918	918
Validation	115	115
Test	115	115
Total	1148	1148

labels are vector line which look like $-\text{[class_id, x_center, y_center, width, height]}$

‘class_id’ shows any object is present or not. It will be 1 if it identify any building and 0 in case of absence of any building.

‘x_center, y_center’ represents the coordinates of the object present in a image.

‘width, height’ are the dimensions of box.

Model

YOLOv8 (You Only Look Once, version 8), developed by Ultralytics, is a state-of-the-art computer vision model used for real-time object detection, segmentation, and classification.

Unlike traditional object detection models that use a two-stage approach—first proposing regions of interest and then classifying them—YOLO treats detection as a single regression problem. It passes the entire image through the convolutional neural network a single time to simultaneously predict bounding box coordinates and class probabilities.

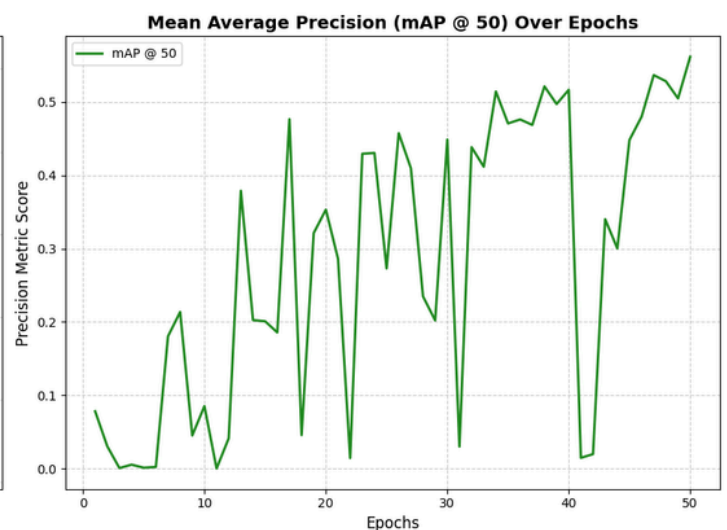
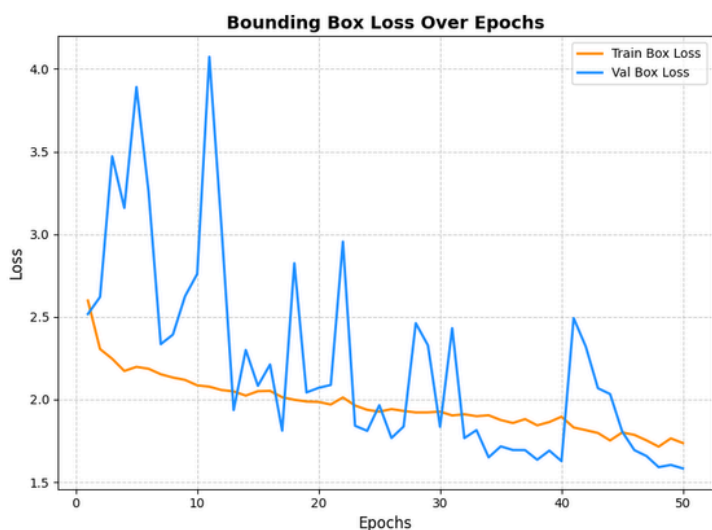
The YOLOv8s (small) variant was selected for this assignment. It contains ~11.1 M parameters. It offers a favourable balance between detection accuracy and computational cost, making it suitable for training on Google Colab's free-tier Tesla T4 GPU within a reasonable time budget.

Training

Title	Values
Epochs	50
Batch Size	16
GPU	NVIDIA Tesla T4
Training Time	~0.201 hours (~12.1 minutes)

Results

Metric	Your Value	What it means in plain English
mAP@50	~0.562	It measures the overall accuracy of your detector on a scale from 0 to 1. A score of 0.56 means that when checking predictions that overlap actual buildings by at least 50%.
Precision	~0.714	Out of all the bounding boxes the model *guessed* were buildings, roughly 62% of them were actually correct buildings.
Recall	~0.545	Out of all the actual buildings physically present in the satellite images, the model successfully detected 51% of them. High recall means the model doesn't overlook or miss real targets.
F1 Score**	~0.618	The harmonic mean (balance score) between Precision and Recall. Because both metrics are evenly matched, your F1 Score indicates a highly stable detector that doesn't skew heavily toward over-predicting or under-predicting boundaries.



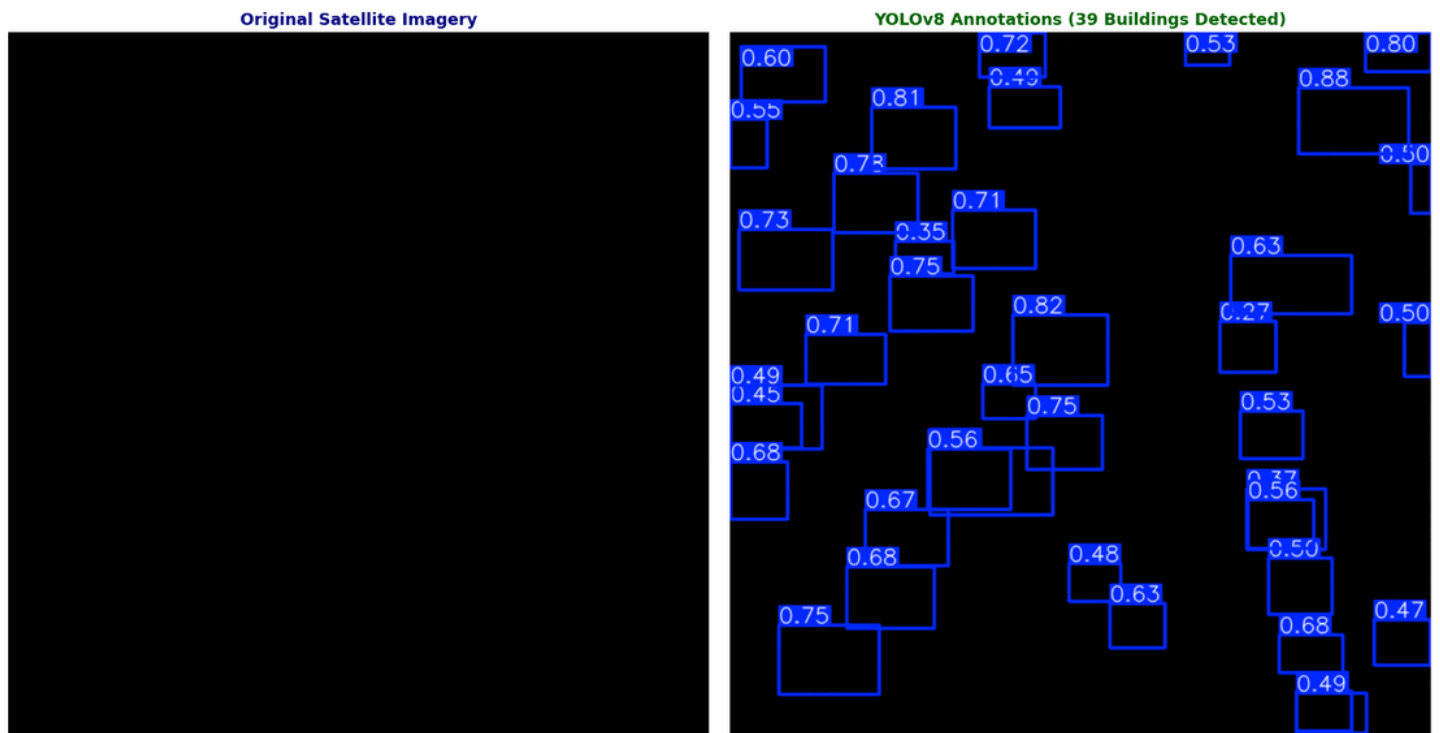


Image: RGB-PanSharpen_AOI_3_Paris_img1010.jpg

Buildings detected : 39

Avg confidence : 0.60

Processing time : 181.4 ms

Observation

The model reached reasonable detection quality within the 50-epoch budget (~12 minutes on T4) with the precision of 71.4%. The data split (80/10/10) was well-structured with perfectly matched image-label pairs across all three splits.

Recall of 54.5% reveals that the model misses nearly half of actual buildings. One image yielded 39 detections at 0.60 confidence, while two others produced very few or zero detections, suggesting the model struggles.

The most impactful improvement would be to use YOLOv8m or YOLOv8l (medium or large) with a higher epoch count (100–150). Additionally, moving from axis-aligned bounding boxes to oriented bounding boxes (OBB) – supported in YOLOv8 .